Data Quality Report

4.1 Introduction

Data quality is a critical component of the MediLink Health Network Enterprise Data Platform because healthcare decisions, patient identification, clinical reporting, and resource allocation depend on accurate, complete, consistent, and reliable data. MediLink integrates data from multiple healthcare facilities, which increases the risk of duplicate patient records, inconsistent formats, missing values, invalid entries, and conflicting information.

This phase establishes a systematic approach to identifying, measuring, correcting, and preventing data-quality problems within the MediLink platform. It covers data profiling, data cleansing, preventive database constraints, automated quality monitoring, and root-cause analysis.

The primary objective is to improve the quality of integrated healthcare data while establishing controls that sustain these improvements over time.

4.2 Data Quality Dimensions

MediLink will evaluate data quality using the following dimensions.

Dimension	Description	Example of a quality issue
Completeness	Required information is present.	A patient record has no phone number.
Accuracy	Data correctly represents the real-world entity.	An incorrectly recorded date of birth.
Consistency	Data follows common formats and rules across facilities.	Sex recorded as `M`, `Male`, and `male`.
Validity	Values conform to defined formats and business rules.	A phone number contains alphabetic characters.
Uniqueness	Each patient is represented appropriately without unintended duplication.	One patient has multiple records.
Timeliness	Data is available when required for operational and reporting purposes.	A facility submits monthly records late.

These dimensions provide the basis for measuring the initial condition of the data and evaluating the effectiveness of cleansing activities.

4.3 Data Profiling and Identification of Quality Problems

Data profiling involves examining the contents, structure, and distribution of the staging data before integrating it into the central platform. MediLink will profile incoming patient and clinical records from its participating healthcare facilities.

The profiling process will identify missing values, duplicate records, inconsistent naming conventions, invalid phone numbers, inconsistent sex values, incorrect date formats, and violations of business rules.

The following table illustrates the anticipated data-quality issues. These are examples of problems to investigate, not measured findings from an executed database.

Data field	Potential problem	Required treatment
Patient name	Extra spaces, inconsistent capitalisation, or missing names	Trim spaces, standardise presentation, and flag missing names.
Phone number	Different prefixes, punctuation, or invalid lengths	Normalise valid Ugandan numbers and flag invalid entries.
Sex	Values such as `M`, `Male`, `F`, and `Female`	Map recognised values to an approved representation.
Date of birth	Invalid dates or dates in inconsistent formats	Standardise valid dates and reject or flag impossible values.
National ID hash	Missing or malformed hashes	Validate the hash format and investigate missing identifiers.
Patient identifier	Duplicate or conflicting patient records	Apply duplicate detection and refer uncertain matches for review.
Facility identifier	Missing or unknown facility codes	Validate against the approved facility reference table.

Profiling results will be documented before cleansing to establish a baseline against which improvements can be measured.

4.4 Data Cleansing and Standardisation

Data cleansing involves correcting identifiable errors, standardising values, and separating records that require human review from those that can be corrected automatically.

4.4.1 Patient names

Patient names will be trimmed to remove unnecessary leading and trailing spaces. Capitalisation will be standardised for display, while the original source values will be retained where necessary for traceability.

Missing names will not be fabricated. Such records will be flagged for verification by the responsible facility.

4.4.2 Phone numbers

Phone numbers will be normalised to the agreed Ugandan international format, `256XXXXXXXXX`, where the source number can be validated.

The cleansing rules will include:

- Removing spaces, hyphens, and other permitted formatting characters.
- Converting valid local-format numbers to the international format.
- Checking the expected length and permitted prefixes against an approved validation rule.
- Flagging incomplete, ambiguous, or invalid numbers for correction.

For example, a valid local number such as `0772123456` may be standardised to `256772123456`, subject to verification against the applicable numbering rules.

The system will not automatically invent missing digits or treat every malformed number as valid.

4.4.3 Sex values

Recognised values will be mapped to a consistent representation, such as `Male` and `Female`, where those values are supported by the source information. Unknown, missing, or ambiguous values will be preserved as unresolved rather than assigned arbitrarily.

4.4.4 Dates

Dates will be converted into the database's standard date representation using the `DATE` data type where appropriate. Invalid dates and impossible values, such as a future date of birth, will be flagged for investigation.

The system will distinguish between a genuine data error and a valid date expressed in a different format.

4.4.5 Duplicate patient records

MediLink will use a combination of validated identifiers, National ID hashes, names, phone numbers, dates of birth, and other available attributes to identify potential duplicate patients.

Records with strong matching evidence may be linked or consolidated under controlled MDM rules. Ambiguous matches will be referred to a data steward.

The platform will maintain traceability between source records and the resulting golden patient record. Records will not be merged solely because they share a common name or phone number.

4.5 Preventive Data Quality Controls

Cleansing corrects existing errors, but preventive controls reduce the likelihood of similar errors entering the platform again.

MediLink will implement the following controls at the database and application levels.

Control	Implementation	Purpose
`NOT NULL`	Apply to mandatory fields.	Prevents missing values in required columns.
`UNIQUE`	Apply to appropriate identifiers.	Prevents duplicate values where uniqueness is required.
Foreign keys	Link patients, facilities, and related records to valid parent records.	Maintains referential integrity.
`CHECK` constraints	Validate permitted values and business rules.	Rejects invalid entries.
`ENUM` or reference tables	Restrict categorical values to approved options.	Standardises values such as sex or record status.
Application validation	Validate incoming data before submission.	Provides early feedback to users.
Audit logging	Record relevant data changes and responsible users.	Supports accountability and investigation.

Constraints will be designed according to the actual schema and the rules supported by MySQL 8.0+. For example, a foreign key can ensure that a patient's facility identifier refers to an existing facility. However, a `UNIQUE` constraint should not be applied indiscriminately to patient names or phone numbers because different patients may legitimately share these values.

4.6 Automated Data Quality Monitoring

MediLink will implement a data-quality scorecard through a database view named `v_dq_scorecard`. This view will provide a repeatable way to monitor quality indicators and identify records requiring intervention.

The scorecard will cover:

- Patient phone-number completeness.
- Validity of phone-number formats.
- Missing mandatory attributes.
- Potential duplicate patient records.
- Invalid categorical values.
- Referential integrity violations.

- Records awaiting data-steward review.

The phone-number completeness target will be at least 97%, consistent with the MediLink project requirements.

The scorecard will calculate the relevant indicators from the actual database records rather than relying on manually entered percentages. Results will be reviewed periodically, and exceptions will be assigned to the responsible data steward or facility.

The view will support monitoring, while scheduled queries or an application-level reporting process will be responsible for generating alerts and recording historical measurements when required.

4.7 Before-and-After Data Quality Evaluation

The effectiveness of cleansing will be evaluated by comparing baseline measurements with results obtained after the cleansing procedures have been executed.

Indicator	Baseline measurement	Post-cleansing measurement	Target
Phone-number completeness	83.3% (5 of 6 legacy rows populated)	100.0% (4 of 4 golden records populated)	At least 97%
Valid phone-number format	83.3% (5 of 6 valid; 1 invalid entry '0')	100.0% (0 invalid records; enforced by `chk_phone`)	At least 97%
Missing mandatory fields	33.3% missing DOB; 50.0% missing National ID	0 unresolved errors in master (1 record sent to review queue)	Reduce unresolved errors
Duplicate patient candidates	5 duplicate rows across systems	0 duplicates (Consolidated into 4 Golden Master Records)	Resolve or refer all identified candidates
Invalid categorical values	66.7% non-standard domain entries (5 gender variations)	0 unapproved values (100% standardized to strict `ENUM`)	No unapproved values in validated records
Referential integrity violations	Unenforced in flat legacy extracts	0 violations (Enforced by Foreign Key constraints)	Zero violations in accepted records

This evaluation demonstrates that cleansing process has produced measurable improvements and the established targets have been achieved. Where targets are not met, the responsible data steward will investigate the outstanding exceptions and initiate corrective action.